

Thermal Phase Shifters with a Focus on Optical Linear Multipliers

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Abstract—This paper addresses the integration of energy-efficient thermal phase shifters into Mach-Zehnder interferometers (MZIs) for optical linear multipliers. The study uses tools such as Lumerical HEAT, MODE, FDTD and INTERCONNECT, along with MATLAB modeling. The study investigates the impact of various heater materials and configurations on phase modulation and optical loss. Key findings include the superior performance of aluminum as a heater material and the significant influence of wire distance on optical efficiency, contributing to advancements in optical neural network (ONN) implementations.

Keywords—thermal phase shifter, Mach-Zehnder Interferometer, MZI, optical linear multipliers, silicon photonics, coherent light, energy efficiency, optical simulation, heater efficiency, optical loss

I. INTRODUCTION

The Implementation of silicon photonics into current data processing systems has significantly improved computational speed and efficiency. Unlike traditional electrical systems, which experience limitations in energy efficiency and heat management as data volumes grow, silicon photonics uses light to analyze and transport information at extraordinary rates. This capacity makes it a promising alternative for meeting the needs of high-bandwidth, energy-efficient computing [5].

Mach-Zehnder Interferometers (MZIs) are crucial components that allow for accurate optical data manipulation. These devices operate by splitting light into two channels, applying phase shifters to one or both, and then recombining them to perform interference-based

activities. This functionality is required for activities such as matrix-vector multiplication, which is a fundamental operation in optical computing and neural networks [6].

The operation of a MZI is heavily reliant on the phase shift imparted to the light in each arm. The phase shift changes the interference pattern at the output, impacting both transmitted and reflected light intensity. Even little phase alterations may end up in considerable changes in output, making phase control a critical component of MZI operation. This sensitivity to phase changes is especially useful in optical systems that require precision, such as optical neural networks (ONNs), which use MZIs as multipliers to conduct linear algebra computations.

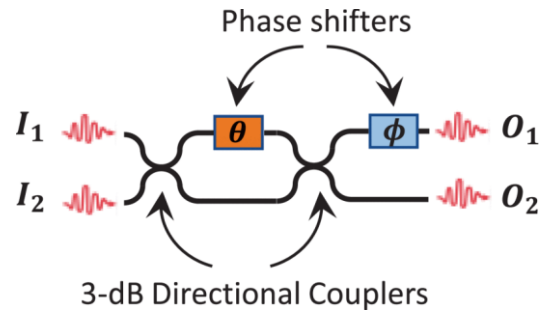


Fig. 1. MZI structure.

Thermal phase shifters are an important component of MZIs, allowing to alter the phase of light by locally heating the waveguide material. This phase control is critical for calibrating and tweaking MZI-based processors so that they run accurately and effectively. However, energy efficiency and scalability of these phase shifters remain issues to be solved, particularly when integrating them into large-scale optical neural networks (ONNs).

Thermal phase shifters are preferred in many applications because they are simple and effective. However, their performance is determined by several elements, including the heater's material, its distance from the waveguide, and the waveguide material's thermal properties. Optimizing these parameters is critical for developing energy-efficient and scalable MZI-based systems, which are required to advance silicon photonic technologies and integrate them into high-performance computing and communication systems. These topics will be covered in greater depth in this paper.

II. RELATED WORKS

A. Impact of Fabrication Process Variations on MZI Neural Networks

Fabrication process differences have a major impact on the performance of MZI-based neural networks. These variances can cause discrepancies in the physical size and qualities of the MZIs, leading to phase errors, reduced optical efficiency, and lower overall system accuracy. MZI meshes rely largely on accurate phase shifts for their operations, therefore even slight manufacturing defects can compound and impair the functionality and energy efficiency of optical neural networks. To achieve consistent performance across several fabrication methods, this problem must be addressed by precise calibration, advanced modelling, and optimization strategies [5].

B. MZI devices can be used for complex-valued neural networks with low power consumption

Mach-Zehnder interferometers (MZIs) improve neural networks by allowing precise manipulation of light to conduct complex-valued arithmetic operations. In neural networks, MZI is utilized to efficiently implement matrix multiplications by utilizing optical signal phase and amplitude control. This enables high speed, parallel data processing while consuming less energy, making it perfect for scaling advanced neural network topologies. Their incorporation into photonic circuits enables faster convergence, better generalization and significant gain in computational efficiency for activities that need complex-valued representation [7].

C. Small phase changes greatly affect light output in MZI devices

Small phase shifts in Mach-Zehnder interferometers (MZIs) have a significant impact on their light output due to the interference principle. When light is split into two paths within the MZI, the relative phase difference between the paths governs how the recombined light interacts. Even modest alterations in this phase difference can dramatically affect the balance of constructive and destructive interference, resulting in considerable changes in the intensity and pattern of the output light.

The phase shifts can be generated by minor variations in temperature, optical path length, or refractive index. While this sensitivity is critical for applications such as sensing and signal processing, it also necessitates precise control to provide consistent and predictable performance. Small, unplanned fluctuations can disturb the expected outcome, emphasizing both the power and challenges of leveraging phase-dependent effects in MZI devices [8].

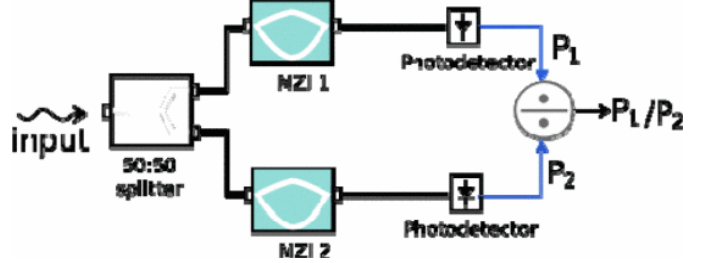


Fig. 2. X-type radiometric wavelength measurement system

D. Training schemes can improve MZI phase shift configuration, increasing optical neural network accuracy

Training systems improve MZI phase shift configurations by correcting mistakes that occur naturally in optical devices. A stepwise training strategy gradually tunes the phase shifts of each MZI. Each phase shifter is carefully adjusted to compensate for mistakes introduced by earlier system components. This strategy not only decreases the buildup of errors, but it also improves the optical neural network's overall performance and accuracy, all without requiring any actual hardware changes [9].

E. Diagonal paths in MZI meshes

Diagonal paths in Mach-Zehnder Interferometer (MZI) meshes are critical for reducing phase error accumulation. These pathways provide independent access to each phase shifter, allowing for accurate calibration and monitoring. Diagonal paths provide direct channels via individual MZIs, facilitating accurate phase adjustments and effectively decreasing the impact of mistakes that can accrue during signal processing

This exact control over phase settings improves time and energy efficiency. Accurate phase setup reduces the need for substantial error correction, resulting in faster processing times. Additionally, limiting error accumulation reduces energy consumption by requiring fewer resources for correction efforts. As a result, including diagonal pathways in MZI meshes not only enhances signal integrity but also optimizes overall system performance [10].

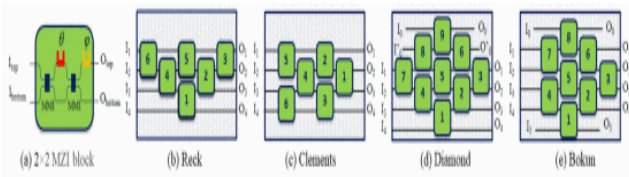


Fig. 3. 2x2 MZI building block with internal and external phase shifters, and 4x4 reconfigurable optical processor with different mesh topologies (b-e) [10].

F. Challenges and Benefits of Thermal Waveguide Tuning

Heating a waveguide using an adjacent metal wire involves passing an electric current through the wire, which generates heat due to electrical resistance. This heat is transferred to the nearby waveguide, causing a localized increase in temperature. Since the refractive index of the waveguide material is temperature-dependent, this thermal effect alters the light propagation characteristics within the waveguide, enabling dynamic control over signal routing.

The metal wire's size and proximity to the waveguide influence its thermal tuning efficiency. Larger wires or those positioned closer to the waveguide can deliver more heat, causing more noticeable changes in the refractive index. However, excessive heating might result in higher optical losses or damage to the waveguide. Thus, optimizing the size and positioning of the heating wire is critical for providing precise and efficient control over light signals, which is required for applications in reconfigurable photonic circuits and optical communication systems [11].

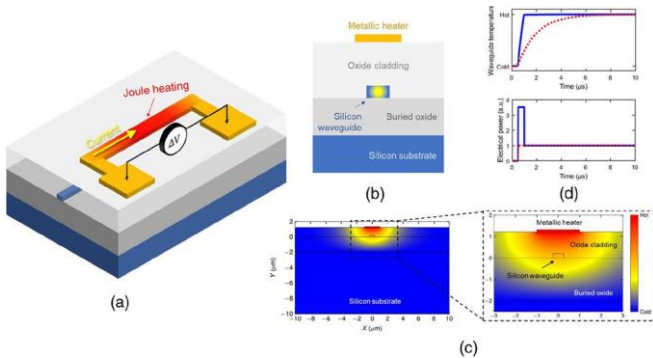


Fig. 4. Illustration of a TOPS using a metallic heater on top of the waveguide. (b) Cross section of the TOPS. (c) Simulated temperature distribution of the TOPS. (d) Temporal response of the TOPS upon a square electrical signal applied to the heater with (solid blue line) and without (dotted red line) employing pulse pre-emphasis. The considered TOPS comprises a $500\text{ nm} \times 220\text{ nm}$ Si waveguide with a $2\text{ }\mu\text{m} \times 100\text{ nm}$ Ti heater on top. The gap between the waveguide and the heater is $1\text{ }\mu\text{m}$. The temperature distribution in the cross section was obtained by solving the conductive heat equation using the COMSOL Multiphysics simulation tool. We considered the thermal constants reported in the literature. 20 A nonuniform tetrahedral mesh, with element sizes ranging from 1 to 500 nm, was employed. A conductive heat flux boundary condition with a

heat transfer coefficient of $5\text{ W}/(\text{m}^2\text{ K})$ was set on the surface. The temperature of the remaining boundaries was fixed at 293.15 K (cold) [11].

This paper also addresses the results of how the metal wire behaves when placed at different distances from the waveguide, as well as how variations in the wire's material affect waveguide loss and efficiency. Simulated results are shown to demonstrate the impact of these parameters, providing useful insights for enhancing waveguide performance for increased efficiency and lower losses

III. METHODOLOGY

For this project we wanted to explore how different heater materials (referred to as the “wire”) affect thermal phase shifter energy efficiency and optical performance. We chose four different materials for our project shown in Table 1.

TABLE I. WIRE MATERIALS

Materials	Thermal Conductivity $[\text{Wm}^{-1}\text{K}^{-1}]$	Refractive Index At 1500nm
Aluminum	237	1.44
Copper	401	0.606
Gold	318	0.558
Silver	406	0.538

These materials were chosen for their common use as conductors to our knowledge. To our current knowledge, aluminum has been used for thermal phase shifting and we wanted to explore and compare it against three other materials that we deemed to be good conductors too.

We also wanted to explore how the distance of this wire from the waveguide affects energy efficiency and optical performance. We chose three distances for the wire location above the waveguide: 200nm, 500nm, and 1μm. The authors in [1] discuss how a waveguide distance of 1μm or more should not introduce any significant optical loss in the waveguide below.

To fully explore thermal phase shifters with respect to optical linear multiplication we decided to divide the project up into three sections: thermal efficiency, optical loss, and a section showcasing optical linear multiplication. This was a necessary decision because different simulation software was needed in each case.

We decided to simulate a strip waveguide in the thermal efficiency and optical loss sections of this project. Mach-Zehnder Interferometers are composed of directional couplers and waveguides. The waveguides are the section where the thermal heater will act on the device resulting in a phase change of the material. Finding

thermal efficiency on just one waveguide will translate to the thermal efficiency of whole mesh for optical multiplication.

A. Thermal Efficiency Simulation

Analyzing the thermal efficiency of a particular thermal phase shifter had to be done in two steps.

The first step is for finding the thermal performance for the phase shifter. The thermal characteristics were analyzed with a simulation software called Ansys HEAT Transfer (HEAT) solver. The main components of this simulation are:

1) Silicon base

The dimensions of the base are 40um tall, by 100um wide, and 1um thick (although this dimension is not important since the simulation is 2D). A boundary condition was placed at the bottom of the base to set the ambient temperature for the simulation to 300K.

2) Cladding

The oxide cladding is 12um tall, by 100um wide, and 1um thick.

3) Air

The air above the wafer was also included for thermal conduction. It was 5.5um tall, 100um wide, and 1um thick. Boundary condition were included for the air and oxide cladding interface so that heat transfer from the wire to the air can be simulated. The values this boundary condition were taken from [2].

4) Silicon waveguide

The silicon waveguide was designed for a guided TE mode. The waveguide is 220nm tall, 450nm tall, and 1um long.

5) Heating element (wire)

The wire dimensions were held constant at 200nm tall, 2um wide, and 1um long. The main changes to this object was its position relative to the waveguide and the material that the wire was to be set to.

6) Uniform power region

The uniform power region is extremely important in our HEAT simulation. It is a region that fills the space of any object in the simulation, which in our case was the wire. The power can be set in the wire to simulated a voltage applied and current heating the wire. The power was changed during the HEAT simulation with the sweep tool and each power value resulted in a temperature grid that the temperature monitor recorded.

7) Temperature monitor

There was a temperature monitor placed exactly on the dimensions of the waveguide. It recorded the temperature throughout the waveguide for every power setting in the wire.

The HEAT simulation can be seen in Fig 5 with a macro view of the base (purple), oxide cladding (light blue), air (green), the waveguide (purple), and the wire (grey).

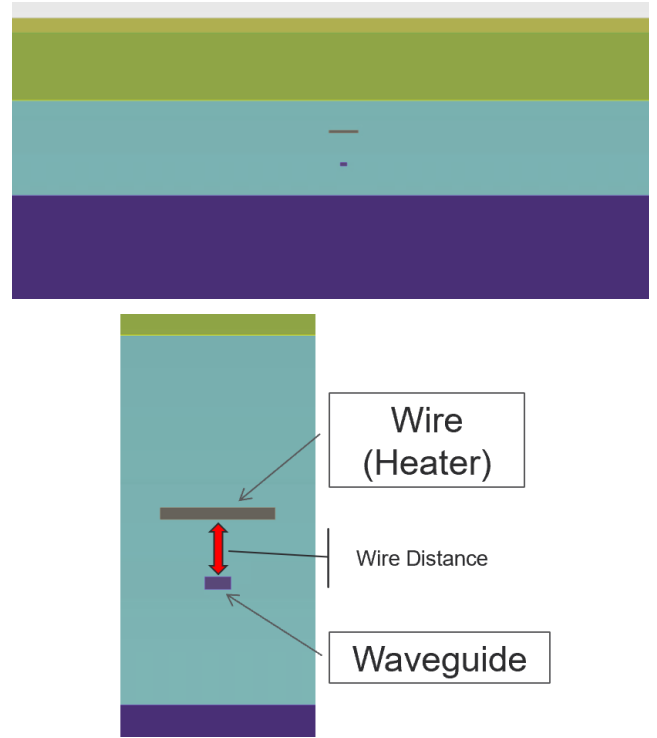


Fig. 5. The HEAT simulation setup.

In Fig 6 the heat dissipation from the wire into its surroundings can be clearly seen. The really dark red rectangle is the wire and the waveguide is engulfed in the red area just below. The heat from the wire is able to affect the waveguide thoroughly. It can also be seen that the heat is traveling upwards to the air where the ambient air temperature was set to 300K. This image also shows the importance of having the waveguide very close to the wire so that the waveguide experiences as much heat as possible from the wire. Having the wire too far away would lead to more heat needed from the wire to change the temperature of the waveguide a meaningful amount.

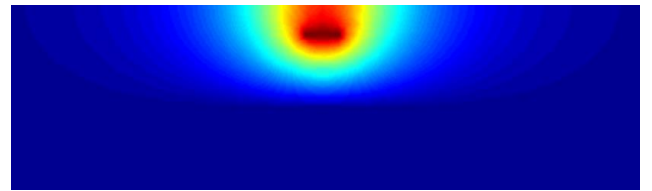


Fig. 6. HEAT dissipation from the wire to its surroundings.

The second step was to transfer the thermal characteristics from Ansys HEAT into Lumerical MODE for an FDE analysis. FDE stands for finite difference eigenmode solver. In [4], FDE is described as a solver that solves Maxwell's equations on a 2D cross section as seen

in Fig 7. It utilizes a mesh that structure in the solver region to get the spatial profile of modes as well as frequency dependence of modes. The outputs of the simulation are effective index, mode field profiles, and more. For our project we focused on observing the effective index change in the waveguide which results in a phase shift on the MZI arm.

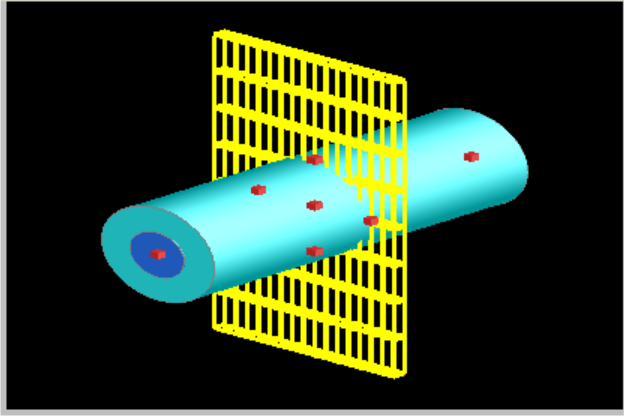


Fig. 7. An FDE cross-section (yellow) with the mesh structure on a 2D plane of an optical device [4].

The main components of the FDE simulation are:

1) *Silicon base*

The same silicon base as from the HEAT simulation with the same dimensions and location. This time the material properties of the silicon are geared towards optical simulations.

2) *Cladding*

The same cladding dimensions and location as it was in HEAT. The material properties now reflect the optical behavior of SiO₂.

3) *Temperature mesh*

The output of the HEAT simulation was a temperature mesh for each power unput simulated into the wire. This temperature mesh contains thermal information based on each power input value. When a certain power is chosen, the mesh updates to contain the temperature values for each mesh cell.

4) *Custom heat dependant silicon*

For this FDE thermal simulation, a custom material was made with silicon properties along with specific thermal behavior. This is called index perturbation. The simulation allows for index perturbation grids to be imported into Lumerical MODE which a custom material can react to. The temperature mesh that was important from HEAT was imported as an index perturbation attribute. The temperature dependence of the refractive index of silicon is $1.87 \times 10^{-4} \text{ K}^{-1}$ at 1500nm. This value was used in the custom material so that the waveguide

temperature mesh would impact the effective index at each mesh cell location in the simulation.

5) *Silicon waveguide*

This is the same silicon waveguide dimensions and location as in the HEAT simulation except with the temperature dependent silicon custom material.

6) *FDE simulation region*

This region was made large enough to encapsulate the 2D cross section of the waveguide.

In Fig 8, the FDE simulation can be seen. The base in shown in red, the cladding in grey, the waveguide in red, the temperature grid in purple, and the FDE mesh for increased accuracy overlayed on the waveguide in orange. The red/green box is where the FDE simulation region is located.

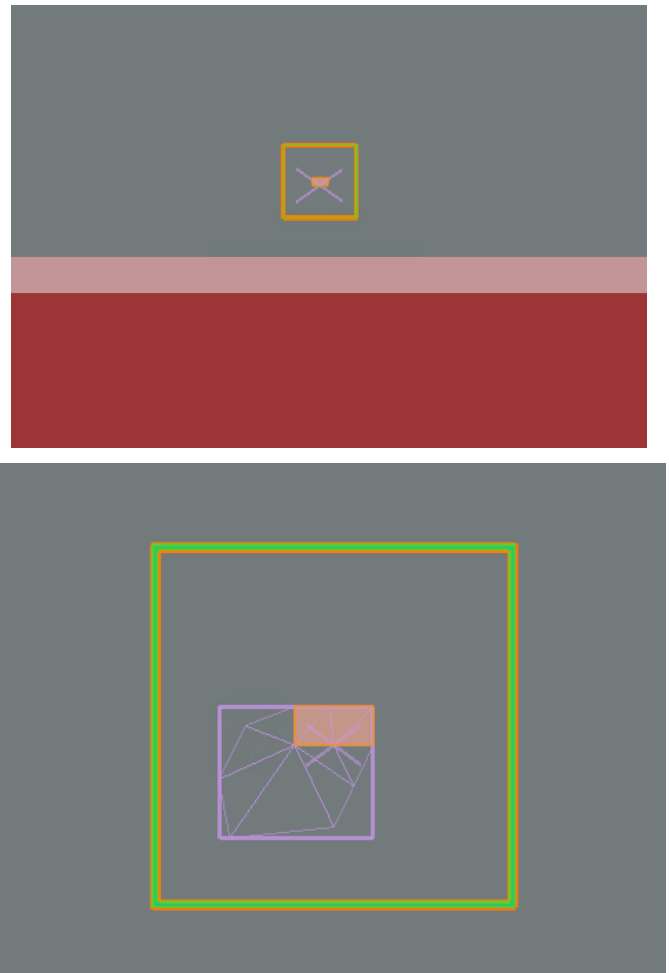


Fig. 8. The FDE simulation setup.

Scripting was used to iteratively loop through the temperature grids generated from the HEAT simulation in the FDE simulations. Each wire material and each wire distance had multiple FDE simulations run to collect the data points for each change in effective index at every new

power input to the thermal heater. The Lumerical scripting code used was influenced by [2] and it is shown in Fig 9.

```
clear;
power = linspace(0.02,0.1,10); # total power in
Watts
neff = matrix(length(power));
# import T map
matlabload("T_Al_1um.mat");
select('temperature');
importdataset(T);

switchtolayout;
# find neff at room temperature
setnamed('temperature','enabled',0);
setanalysis('wavelength',1.55e-6);
findmodes;
neff0 = real(getdata('mode1','neff'));

# read Temperature Mesh
for (i=1:length(power)){
switchtolayout;
setnamed('temperature','enabled',1);
setnamed('temperature','P_index',i);
findmodes;
neff(i) = real(getdata('mode1','neff'));
}

L = 20e-6;
lambda = 1.55e-6;
phase = 2*pi*(neff-neff0)*L/lambda;
plot(power, phase, 'power (W)', 'phase');
```

Fig. 9. Lumerical script for iterating through temperature grids, running the FDE simulation to obtain the new effective index, and plotting the phase shift with respect to power.

The output of the second step is a plot that shows the phase change for increasing power of the thermal phase shifter based on the change in the effective index of the material as temperature is increased. The equation for a change in phase due to a change in the effective index is:

$$\Delta\phi = \frac{2\pi(n_{eff_{end}} - n_{eff_{start}})L}{\lambda}.$$

The common metric for efficiency regarding a thermally tuned waveguide is [mw/FSR] as discussed in [1]. This metric is answering the question of how much power is needed to shift the phase of the optical signal by 2π .

B. Optical Loss Simulation

To perform optical loss simulations we used the Lumerical FDTD optical simulation tool. This simulation tool allows for a 3D simulation of an optical device.

FDTD stands for finite-difference time-domain and is a method used to solve full-wave 3D electromagnetic problems [3]. This solver is extremely useful in modeling

photonics devices as we are in this project. The article in [3] discusses how the FDTD simulation is bound by a simulation region in which the geometry is defined by a mesh. Maxwell's equations are used to solve for the electric field at each mesh cell location. These calculations are done over many steps of time to show the changes in the electromagnetic fields over time. Finer meshes allow for more accuracy but greatly increase computation time and memory requirements.

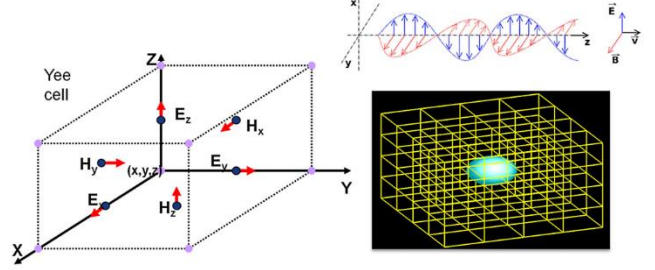


Fig. 10. A mesh cell structure in FDTD simulations [3]

The main components of this simulation are:

1) Optical waveguide

The waveguide's dimensions were kept constant along with its physical position relative to the wire, the same as in the previous simulations except the length of the waveguide matched the wire's length.

2) Heating element (wire)

The wire dimensions were kept the same as in the previous simulations except this time the length was set to 20um. Ideal simulation conditions would have been with a wire and waveguide lengths of 100um or even 200um. This was not practical for our project as we had to perform an FDTD simulation for every material and wire distance combination. Therefore 20um was chosen as a suitable value. We assume that the results we collected would still represent general trends. At 20um the loss may not be as present as it would have been with 200um but the trends amongst materials and wire distances can still be seen.

3) Cladding

The cladding dimensions in this simulation were set to fully cover the simulation region and the devices. This simulation focused on the interaction between the wire and the waveguide and therefore cladding needed to be present but not air or the silicon base.

4) Light source

The light source was coherent and kept at 1550nm for all the simulation runs. It was injected into the center of the waveguide on the left within the simulation region.

5) FDTD solver region

The solver region covered 20um of the wire and waveguide.

6) Mesh grid

The mesh grid was added to help improve the accuracy of the interaction between the wire and the waveguide.

7) Transmission monitors

The transmission monitor placement was crucial. One monitor was placed right after the light source as to serve the purpose of collecting the transmission power when no loss was present. The other transmission monitor was placed at the end of the 20um length to collect the transmission power after optical loss took place due to the presence of the wire in proximity to the waveguide.

In Fig 11, the FDTD simulation can be seen.

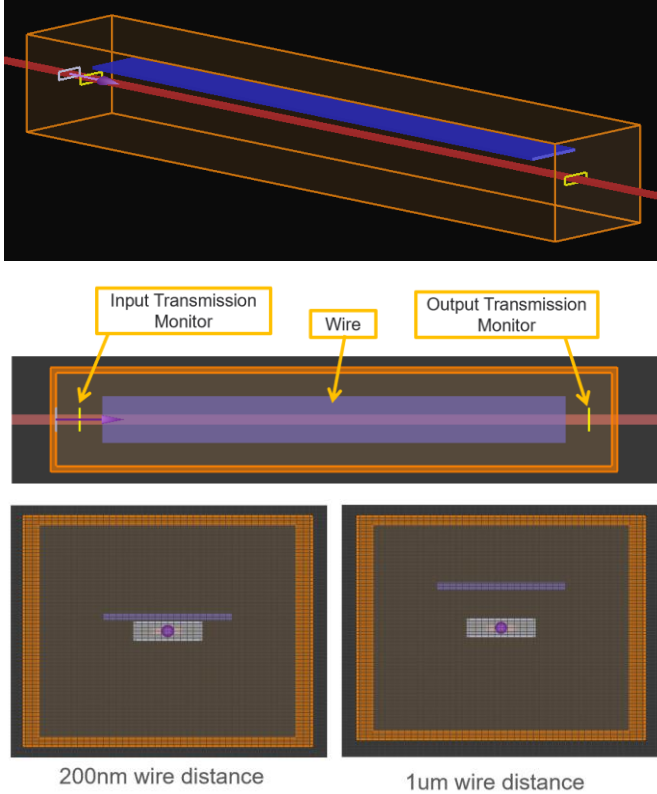


Fig. 11. FDTD simulation setup with the optical transmission monitors shown, the wire, the waveguide, and the 3D simulation region.

To calculate the optical loss, a simple equation was used that subtracted output power as measured by the second power monitor from the input power as measured by the first power monitor.

$$\text{Optical loss} = \text{Output power} - \text{Input power}$$

The Lumerical FDTD simulation did not calculate optical loss due to any structural defects like back reflection or sidewall scattering. Therefore all optical loss values were due to the interaction with the wire which was ideal for this project.

C. Optical Linear Multiplication

Optical linear multiplication all starts with the MZI in our project. We focused on multiplication within one MZI which is a 2x2 multiplication. In general, multiple MZIs are optically connected together to create a mesh structure that represent larger multiplication needs such as for optical neural networks. In Fig 12, the MZI transfer matrix we wish to implement is shown.

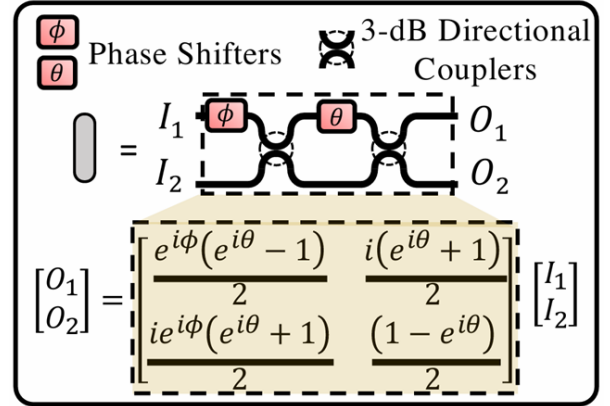


Fig. 12. MZI transfer matrix [12].

In our project we aim to show how optical linear multiplication can be done with a compact model as well as a circuit level model in Lumerical INTERCONNECT. Then we will show how optical loss can affect optical linear multiplication negatively.

First, optical linear multiplication can be done with a compact model, we chose to use MATLAB for this. The MZI transfer matrix is used represent the MZI. The outputs can be obtained through the transfer matrix with the desired inputs as show in Fig 13.

```
phi = 0;
theta = pi/4;
T_MZI = [(exp(1i*phi)*(exp(1i*theta) - 1))/2, (1i*(exp(1i*theta) + 1))/2; ...
          (1i*exp(1i*phi)*(exp(1i*theta) + 1))/2, (1 - exp(1i*theta))/2];
I = [1; 0];
O = T_MZI*I;
```

Fig. 13. MATLAB code for an MZI compact model.

Optical linear multiplication can also be shown with INTERCONNECT. This software allows for rapid high level optical circuit testing by connecting individual optical components together. The software contains the s-parameters of each component of an MZI such as phase shifters or directional couplers which allows for proper behavior to be modeled. In Fig 14, the circuit level model of an MZI can be seen.

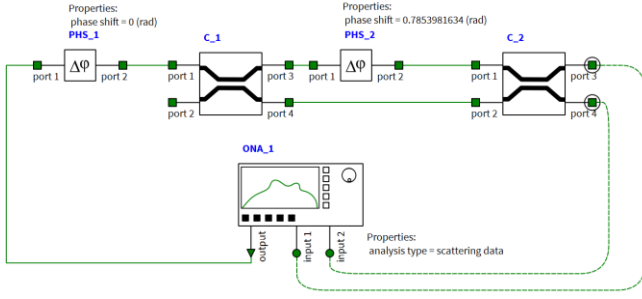


Fig. 14. INTERCONNECT circuit level model of an MZI.

IV. RESULTS

The following sections cover the results of our project.

A. Thermal Efficiency Results

A total of three plots are shown (Fig 15, 16, and 17), each plot is for a different wire distance from the waveguide. In each plot, the four materials are shown. The plots have input thermal phase shifter on the x-axis and the phase shift on the y-axis. There is a horizontal line at 2π to show where each material crossed the FSR phase shift. Materials that cross the FSR phase shift at lower powers have a higher thermal efficiency.

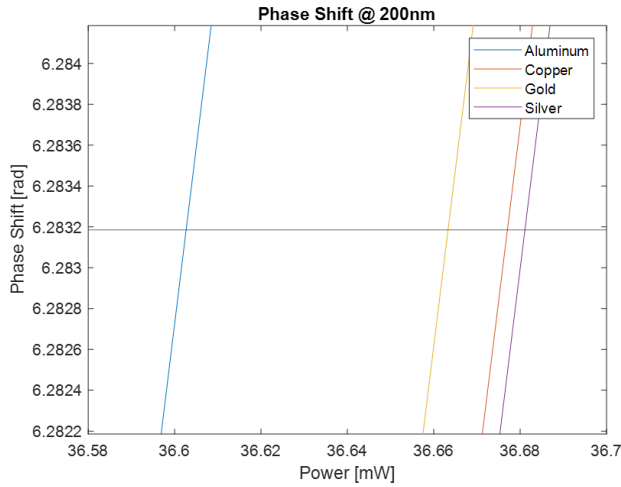


Fig. 15. Thermal efficiency of each material at 200nm wire distance from the waveguide.

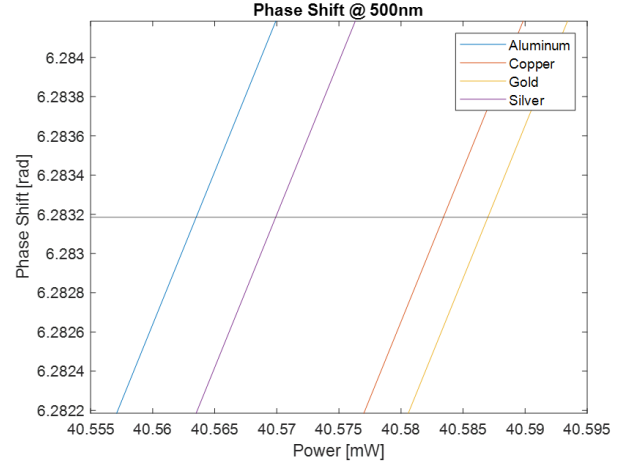


Fig. 16. Thermal efficiency of each material at 500nm wire distance from the waveguide.

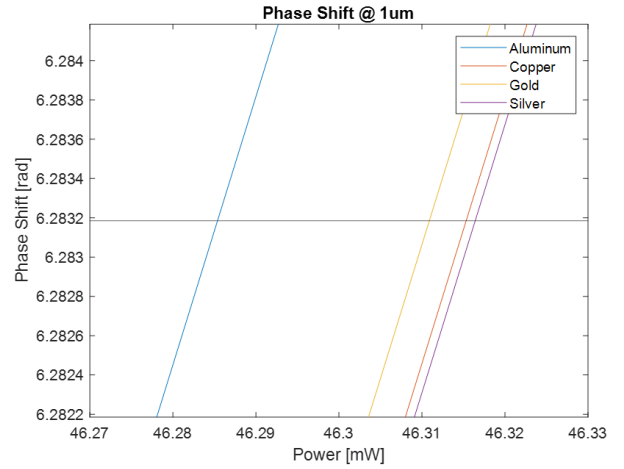


Fig. 17. Thermal efficiency of each material at 1um wire distance from the waveguide.

The plots are summarized in Table II showing the thermal efficiency metric of mw/FSR.

TABLE II. THERMAL EFFICIENCY

Wire Distance [nm]	Phase Shifter Efficiency [mw/FSR]			
	Aluminum	Copper	Gold	Silver
200	36.603	36.677	36.663	36.681
500	40.564	40.583	40.587	40.570
1000	46.285	46.315	46.311	46.317

As we can see, aluminum performed the best across all three wire distances. Silver was the worst material at 200nm and 1um. Gold was the worst performing material at 500nm.

B. Optical Loss Results

Similarly to the thermal efficiency results, three plots are shown (Fig 18, 19, 20) with each plot representing a different wire distance. Each plot contains all four materials. The x-axis represents different wavelengths and the y-axis shows the optical loss at each of those wavelengths.

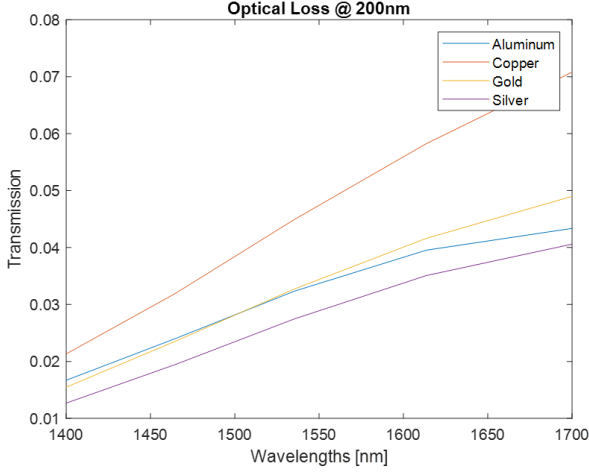


Fig. 18. Optical loss for each material at 200nm wire distance.

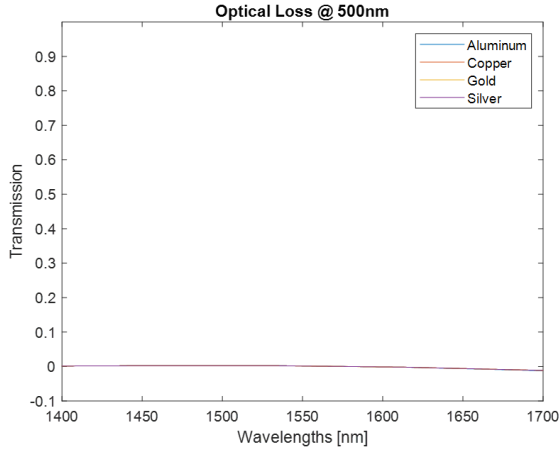


Fig. 19. Optical loss for each material at 500nm wire distance.

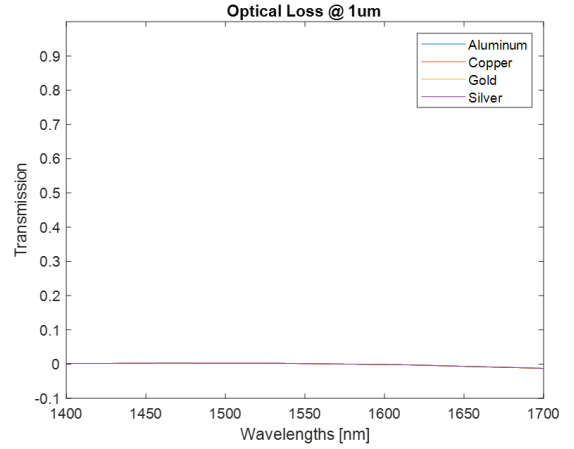


Fig. 20. Optical loss for each material at 1um wire distance.

The results show that at 200nm wire distance Silver performed the best, followed by aluminum, gold, and lastly copper. Loss was not present at 500nm although it was expected. This could be due to the FDTD simulation mesh not being small enough or the MZI length not being long enough for loss to accumulate to a meaningful value. The results for 1um were expected since there should be almost no loss present at 1-2um wire distance [1].

C. Optical Linear Multiplication Results

To prove to ourselves that the MATLAB compact model and the circuit level INTERCONNECT model were the same we ran simulations for each where we kept phi at 0 rad and theta was increment from 0 to π in increments of $\pi/4$. The MZI outputs were obtained and their absolute value was taken. For these simulations only one input was on for the MZI. This was because the spectrum analyzer in the software could only power one input at a time. We obtained the same values in each case shown in Table III.

TABLE III. MZI OUTPUTS

MZI Output	Theta				
	0	$\pi/4$	$\pi/2$	$3\pi/4$	π
O1	0	0.383	0.707	0.924	1
O2	1	0.924	0.707	0.383	0

Now, to show how optical loss can impact optical linear multiplication we used INTERCONNECT and modified the MZI to include optical loss. With phi still at 0 rad we iterated through the same theta values and found that a theta value of $\pi/4$ can have the same absolute value as a $\pi/2$ theta with optical loss in the system, specifically 5.32dB insertion loss as shown in Table IV.

TABLE IV. MZI OUTPUTS WITH LOSS

MZI Outputs	Theta		
	$\pi/4$	$\pi/2$	$\pi/2$ with loss
O1	0.383	0.707	0.383

Optical linear multiplication uses the phi phase shift on the input to help encode input values to the multiplication meshes. The same optical loss is more difficult to be shown with phi involved. When the phi value is changed the absolute value of the MZI is not altered when only one input is on (Table V). This is because the phi value rotates the phase of the input while the theta value dictates the interaction of the light between the two arms which can result in destructive or constructive interference with coherent light.

TABLE V. MZI OUTPUTS WITH PHI

MZI Output	Phi (Theta constant at $\pi/4$)				
	0	$\pi/4$	$\pi/2$	$3\pi/4$	π
O1	0.383	0.383	0.383	0.383	0.383
O2	0.924	0.924	0.924	0.924	0.924

For loss to affect the MZI with the phi phase shift the other MZI input must be present. This way loss in one phase shift can scale the values to appear as a different phi phase shift. This can lead to the same negative effects as show for the theta values with loss.

V. CONCLUSION

Overall, the project was successful. We collected data for thermal efficiency for various materials and how those various materials can impact optical loss. We found that although aluminum was the favorite material for thermal efficiency, it was not the material with the lowest optical loss. Silver actually had the lowest optical loss. Our results show that design decisions such as wire heater positioning within an MZI can greatly impact overall performance. Various software tools from heat simulation to optical simulations with FDE and FDTD were performed showing the great benefit of these tools and they can expedite design decisions. This project was important to us because the field of silicon photonics is growing at a rapid pace and there are key challenges that if solved can truly accelerate progress. We identified thermal energy efficiency of phase shifters and their resulting optical loss to be one of the important designs to get right in this field. These phase shifters are used in mass quantities to perform optical linear multiplication that can be used for accelerating artificial intelligence training and other various matrix-vector multiplication problems. We are excited to have worked on solving one of the many challenging and exciting problems in silicon photonics.

VI. MEMBER CONTRIBUTIONS

Both of us worked to understand all aspects of the project before splitting off and completing individual tasks. This ensured that we both agreed on specific steps to make sure our simulations were done right and the overall story remained constant. We agreed on wire distances, materials, and MZI structure. Mukesh mainly worked on the HEAT simulations, collecting data on thermal efficiency. Kevin mainly worked on the optical loss simulations. We both analyzed the data together for each simulation. Then we worked together to understand and show how optical linear multiplication works with a compact model and the circuit level model. Overall, the roles were split fairly equally between us.

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